

Scalability & Future Plan

Date	04 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The system uses a synthetic dataset instead of real banking data.	Prediction accuracy may differ from real-world scenarios.	High
2	The application is deployed on a single server and cannot efficiently handle a large number of users simultaneously.	Performance may decrease during heavy traffic.	High
3	Prediction history is stored only for the current session and is not saved permanently in a database.	Users lose previous prediction records after the session ends.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited concurrent users on a local Flask server.	Deploy the application on cloud platforms with load balancing and auto-scaling.
2	Data Storage	Uses session storage without a permanent database.	Integrate MySQL or PostgreSQL for secure and permanent data storage.
3	Performance	Model is loaded locally with basic preprocessing.	Optimize preprocessing, use model caching, and improve response time through server optimization.

4	Security	Basic input validation is implemented.	Add user authentication, HTTPS encryption, secure password storage, and role-based access control.
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Integrate a real-world banking dataset and retrain the machine learning model.	3 Month	Improves prediction accuracy and reliability.
Phase 2	Add user login, registration, and prediction history management using a database.	6 Month	Provides personalized and secure user experience.
Phase 3	Implement Explainable AI (XAI) to show the reasons behind approval or rejection decisions with dashboards and analytics.	9 Month	Increases transparency and helps users understand predictions.
Phase 4	Deploy the system on the cloud and develop REST APIs and a mobile application for wider accessibility.	12 Month	Enables large-scale deployment and access from multiple devices.